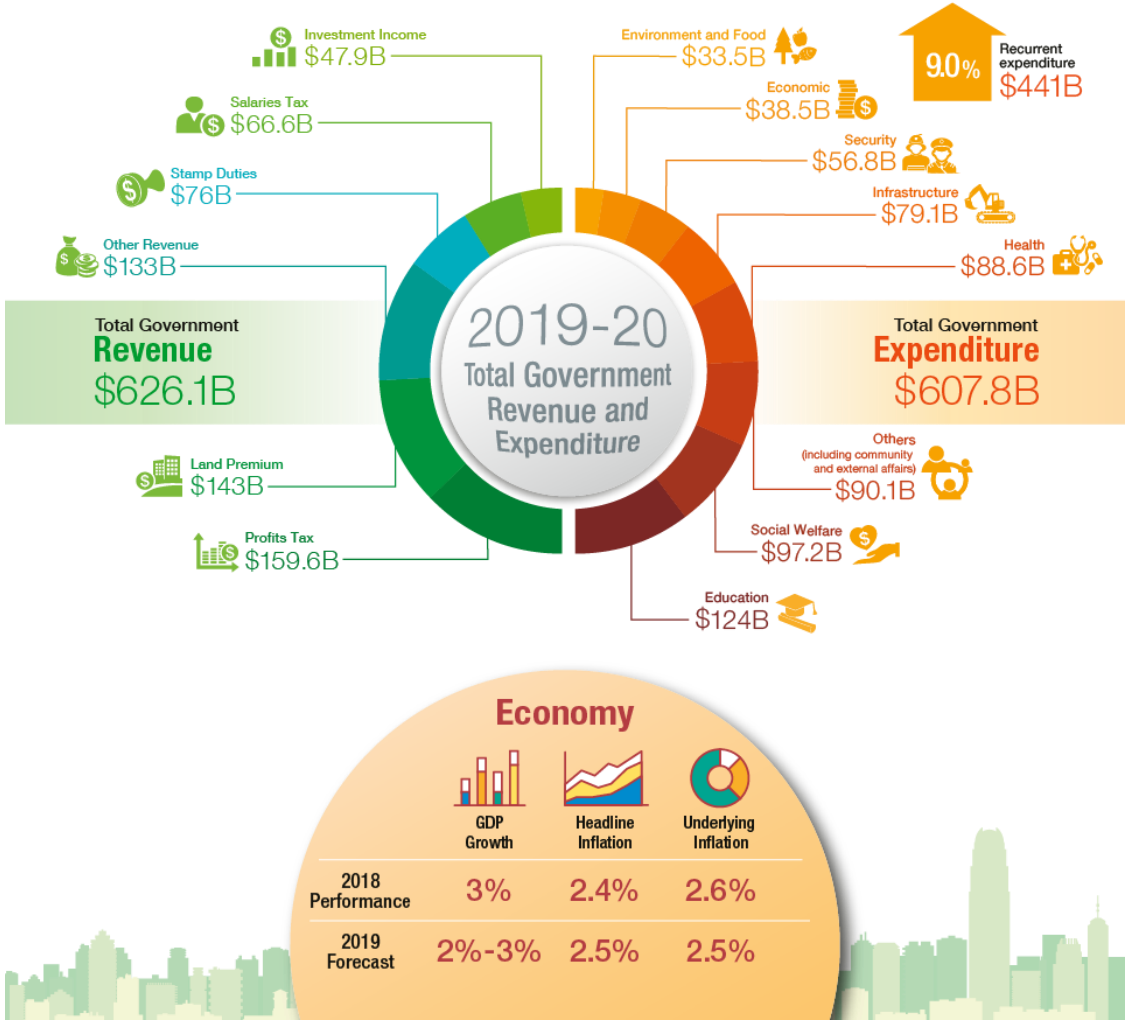
ECLT5930B/SEEM5740B Engineering Economics and Cost Management

Chapter 7
Depreciation and Income Taxes

Huifu Xu
Dept. of Systems Engineering & Engineering Management
The Chinese University of Hong Kong

Importance of tax to government



Tax may affect net income

- Business activities are subject to government tax
  - Personal income is also subject to tax
- After-tax cash flow (ATCF)
  - Changes in the PW, FW, AW, IRR, etc.
- Depreciation plays an important role in tax calculation. By affecting tax payments, depreciation affects the after-tax cash flows.
- We will discuss
  - depreciation;
  - tax calculation;
  - after-tax economic analysis.

Depreciation

- Depreciation is the decrease of value of an asset with the passage of time and use
  - Accounting concept
  - The value of a capital investment should be recovered gradually over years
- Purposes
  - To evaluate the performance of the project or company
  - To prepare capital for future business
  - To calculate tax deduction
    - It determines the value of “taxable income” for a business corporation

Depreciation

- Market value vs. book value:
  - Market value is the value of the asset on the second-hand market.
  - Book value is the value of the asset on the accounting book.
  - Market value and book value can be very different. For IT equipment, its market value is often lower than the book value due to rapidly changing technology.
- Depreciation is an accounting concept that reflects the loss in an asset's value due to the effect of time and use.

Depreciation

- Suppose at the beginning of year 2013, the book value of an asset is $1000. The depreciation amount is $200 for year 2013. At the end of year 2013, the book value becomes 1000 - 200 = $800.
- The $200 depreciation amount reflects the loss of the asset's book value for year 2013.
- Depreciation is tax deductible. In the previous example, $200 will be deducted from the taxable income in 2013.
- Hence depreciation looks like a "cost", as in calculating the taxable income, any costs will be deducted (only the profits are taxable).
- Depreciation is a book cost (there is no actual cash outflow for it).

Which assets can be depreciated

- Depreciation is not allowed for all assets. A depreciable asset must:
  - be used for business purpose;
  - have a determinable useful life which is longer than 1 year;
  - decay, get used up, wear out, become obsolete, or lose value from natural causes;
  - not be inventory or investment asset.

Classification of assets

- Tangible (can be seen or touched)
  - real property: land and generally anything that is growing on, built upon, or attached to land such as buildings, etc.
  - personal property: include machinery, vehicles, equipment, furniture etc. Anything that is tangible excluding real assets.
- Intangible (patents, copyrights, trademarks, etc.)
- Most tangible assets can be depreciated. An important exception is land (determinable life?, lose in value over time?).
- In engineering economic analysis we will focus on tangible assets.

Example 7.1

- Consider a local Pizza business. Which of the following can be depreciated?
  - (1) A baking oven.
  - (2) Pizza dough and toppings.
  - (3) A leased delivery van.
  - (4) A new car that is solely used by the business owner for personal travel.

Example 7.1

- Solution:
  - (1) A baking oven.
    - Yes.
  - (2) Pizza dough and toppings.
    - No. Useful life not longer than 1 year.
  - (3) A leased delivery van.
    - No. Only the owner of the van can claim depreciation from it.
  - (4) A new car that is solely used by the business owner for personal travel.
    - No. Not used for business purpose.

When depreciation starts and when it ends

- A company can begin to depreciate an asset it owns when the asset is placed in service for use in the business.
- Property is considered to be placed in service when it is ready and available for a specific use, even if it is not actually used yet.
- Depreciation stops when the cost of placing an asset in service has been recovered according to the accounting rule or when the asset is sold, whichever occurs first.

Depreciation terminology

- Cost basis (B)
  - Initial cost of the asset, to be allocated to the following years
- Depreciation deduction (d_k)
  - The decrease value of the asset for each year k
- Book value
  - The worth of an asset for each year k shown on the accounting record
  - $(\text{Book value})_k$=cost basis–(total depreciation from Year 1 to k)
- Useful life
  - The number of years over which an asset is depreciated

Depreciation terminology

- Market value (salvage value, SV)
  - The price at which an asset is sold after being used for years
- Recovery period:
  - The number of years over which the cost basis of a property is recovered through the accounting process.
- Recovery rate:

$$\text{Recovery rate in year k} = \frac{\text{depreciation in year k}}{\text{total depreciation amount over the full recovery period}}.$$

Depreciation methods

- Different from country to country, time to time
  - Laws, regulations,…
- Depending on asset types
  - Asset of property life, Recovery period, Calculation method
- Classical methods:
  - Straight-Line (SL), Declining-Balance (DB), DB with switch-over to SL, and Sum of the Year Digits (SYD).
- Current in use in the USA:
  - Modified Accelerated Cost Recovery System (MACRS). The depreciation system in Hong Kong is similar.

Straight-Line (SL) method

- Depreciation deduction is constant each year over the useful life of an asset.

$$d_k = d = \frac{B - SV}{N}$$
$$d_k^* = k \times d$$
$$BV_k = B - d_k^* = B - k \times d$$
$$BV_N = SV$$

  - N = depreciable life of the asset in years (useful life);
  - B = adjusted cost basis;
  - SV = estimated asset value at the end of year N;
  - d_k*= cumulative depreciation through year k:
- Remark: To use this method we must first estimate SV. The actual terminal MV may be different from SV.

Example 7.2

- An asset has an adjusted cost basis of $25,000 with a $2,000 salvage value at the end of its depreciable life of 8 years.

EOY k	d_k	BV_k
0	-	$25,000
1	2,875	22,125
2	2,875	19,250
3	2,875	16,375
4	2,875	13,500
5	2,875	10,625
6	2,875	7,750
7	2,875	4,875
8	2,875	2,000

$$d = \frac{25{,}000-2{,}000}{8} = 2,875,\ BV_k = B - kd.$$

Declining-Balance (DB) Method

- Annual depreciation is a fixed percentage, *R*, of the book value of the previous year
  - R=2/N, 200% DB
  - R=1.5/N, 150% DB

$$d_1 = B(R)$$

$$d_k = B(1-R)^{k-1}(R)$$

$$d_k^* = B[1-(1-R)^k]$$

$$BV_k = B(1-R)^k$$

Example 7.3

- Example
  - Cost basis *B*=4,000
  - Depreciable life *N*=10
  - Estimated SV=0
  - *R*=2/N (200%DB)

EOY k	d_k	BV_k
0	-	$4,000
1	800	3,200
2	640	2,560
3	512	2,048
4	409.6	1,638.4
	...	
10	107.37	429.5

$$d_1 = B(R) = 4000(0.2) = 800$$

$$d_6 = B(1-R)^{k-1}(R) = 4000(1-0.2)^5(0.2) = 262.14$$

$$d_6^* = B[1-(1-R)^k] = 4000[1-(1-0.2)^6] = 2951.42$$

$$BV_6 = B(1-R)^k = 4000(1-0.2)^6 = 1048.58$$

Tax types

- Income taxes
  - assessed as a function of gross revenues minus allowable deductions.
- Property taxes
  - assessed as a function of the value of property owned.
- Sales taxes
  - assessed on the basis of purchases of goods or services and are thus independent of gross income or profits.
- Excise taxes
  - assessed as a function of the sale of certain goods or services often considered nonnecessities and are hence independent of the income or profit of a business.

Income tax rate

- Income tax usually has different tax rate for different intervals (progressive rate)
  - First $10,000: tax rate=5%
  - Next $20,000: tax rate=8%
  - All above: tax rate=10%
  - For income=$31,000, the payable tax is

    10000*5%+20000*8%+(31000−30000)*10%
- This also applies to your salary income

How Salary Tax is Determined

- Taxable salary = Salary – deductions/allowance
- For example:
  - Basic allowance: HK$100,000 for each tax payer
  - Child(ren): HK$50,000 for each child
  - Qualified dependents
  - Donations
  - Home loan interests
  - …

Salary Tax Rate in Hong Kong, Year of Assessment 2020/21 (Progressive Rate)

	Net Chargeable Income	Rate	Tax
	$		$
On the First	50,000	2%	1,000
On the Next	50,000	6%	3,000
	100,000		4,000
On the Next	50,000	10%	5,000
	150,000		9,000
On the Next	50,000	14%	7,000
	200,000		16,000
Remainder		17%	

Example: Taxable salary=450,000

Tax=(450,000-200,000)*0.17+16,000 = 58,500

Separate or Joint Assessment?

- Mr. Wong has a salary of HK$600,000 in year 2020-2021. Mrs. Wong has a salary of X. They have one child.
- If X=HK$90,000
  - Under separate assessment
    - Mr. Wong: taxable salary=450,000→tax=HK$58,500
    - Mrs. Wong: taxable salary = 0 → tax=HK$0
    - Total tax = HK$58,500
  - Under joint assessment (as a family)
    - Taxable salary = 440,000→tax=HK$56,800
- If X=110,000
  - Under separate assessment
    - Mrs. Wong's taxable salary = 10,000 → tax=HK$200
    - Total tax=HK$58,700
  - Under joint assessment
    - Taxable salary = 460,000→tax=HK$60,200

Effective and Marginal Tax Rate

- Sometimes we are interested in the marginal change of tax for a given interval
  - Telling us the effect to the payable tax when changing income, depreciation…
  - For Hong Kong Salaries Tax Rate
    - The marginal income rate for [150000,200000] is 14%
- Sometimes we use the effective tax rate
  - Average tax rate
  - An approximation used to estimate income tax

    Income Tax = taxable income * effective tax rate

Income taxes

- Taxable Income (TI) = Gross income - all expenses (except capital investments) - depreciation deductions.
- The effect of capital investments is considered through depreciation deductions.
- You can think of depreciation as way to spread the capital investment cost over years.

Example 7.4

- A company generated $200M of gross income from sales and incurred $140M of operating expenses this year. A capital investment of $1.5M was made to purchase a specialized equipment at the beginning of the year, and the company deducted $0.1M as the annual depreciation charge for this equipment. What is the taxable income of the company?

- TI = $200M - $140M - $0.1M = $59.9M.

Example 7.5

- Suppose that a firm for a tax year has a gross income of $5,270,000, expenses (excluding capital) of $2,927,500, and depreciation deductions of $1,874,300. What would be its taxable income and federal income tax for the tax year?

If Taxable Income Is:		The Tax Is:	
Over	But Not Over		of the Amount Over
0	$50,000	15%	0
$50,000	75,000	$7,500 + 25%	$50,000
75,000	100,000	13,750 + 34%	75,000
100,000	335,000	22,250 + 39%	100,000
335,000	10,000,000	113,900 + 34%	335,000
10,000,000	15,000,000	3,400,000 + 35%	10,000,000
15,000,000	18,333,333	5,150,000 + 38%	15,000,000
18,333,333	.....	35%	0

Source: Tax Information on Corporations, IRS Publication 542, 2016.

Example 7.5

- Solution
  - Taxable income = Gross income − Expenses − Depreciation deductions = $5,270,000 − $2,927,500 − $1,874,300 = $468,200
  - Income tax = 15% of first $50,000 $7,500

    + 25% of the next $25,000 6,250

    + 34% of the next $25,000 8,500

    + 39% of the next $235,000 91,650

    + 34% of the next $133,200 45,288

    Total $159,188

If Taxable Income Is:		The Tax Is:	
Over	But Not Over		of the Amount Over
0	$50,000	15%	0
$50,000	75,000	$7,500 + 25%	$50,000
75,000	100,000	13,750 + 34%	75,000
100,000	335,000	22,250 + 39%	100,000
335,000	10,000,000	113,900 + 34%	335,000
10,000,000	15,000,000	3,400,000 + 35%	10,000,000
15,000,000	18,333,333	5,150,000 + 38%	15,000,000
18,333,333	. . . . .	35%	0

Source: Tax Information on Corporations, IRS Publication 542, 2016.

Gain (loss) on the disposal of an asset

- When a depreciable asset is disposed (sold), the MV is seldom equal to its BV. Suppose the asset is disposed at year N.
- Gain (loss) = $MV_N - BV_N$
  - If $MV_N > BV_N$, gain on disposal, tax must be paid on the depreciation recapture.
  - If $MV_N < BV_N$, loss on disposal, and it results in earning tax credit.
- Remark: In general, the tax rate for asset disposal is different from the income tax rate.

Example 7.6

- A company sold a piece of equipment during the current tax year for $78,600. The accounting records show that its cost basis is $190,000 and the accumulated depreciation is $139,200. Assume that the effective income tax rate is 40%.What is
  - (a) the gain (loss) on disposal,
  - (b) the tax liability (or credit) resulting from this sale,
  - (c) the tax liability (or credit) if the accumulated depreciation was $92,400 instead of $139,200?

Example 7.6

- Solution
  - (a) The BV at the time of sale is 190,000–139,200 = 50,800.

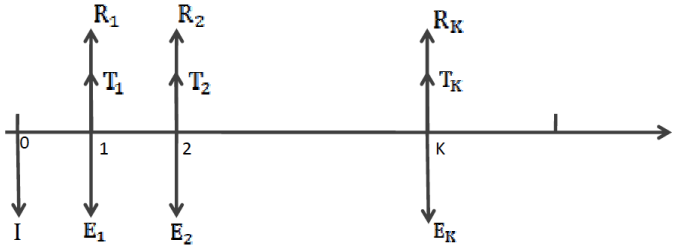
    Hence the gain on disposal is 78,600–50,800 = 27,800.
  - (b) The tax liability is 27800×40%=11,120.
  - (c) The BV at the time of sale is 190,000–92,400 = 97,600.

    Hence the loss on disposal is 97,600–78,600 = 19,000.
    The tax credit is

After-Tax Cash Flow

- Notation
  - R_k: revenue at year k (cash inflow)
  - E_k: expense at year k (cash outflow)
  - d_k: depreciation at year k (treated as outflow here, but not a real cash flow)
  - t: effective tax rate (assumed to be constant)
  - T_k: income tax for year k
  - BTCF_k: Before-tax cash flow in year k.
  - ATCF_k: after-tax cash flow for year k

After-Tax Cash Flow



- Equation
  - $BTCF_k = R_k - E_k$
    - *When $R_k \geq (E_k + d_k)$, tax liability occurs. When $R_k < (E_k + d_k)$, tax credit occurs*
  - $T_k = -t\,(R_k - E_k - d_k)$
  - $ATCF_k = BTCF_k + T_k = (R_k - E_k) - t\,(R_k - E_k - d_k)$

$$= (1-t)(R_k - E_k) + t\,d_k$$

Year	(A) BTCF	(B) Depreciation	(C)= (A)-(B) Taxable Income	(D)=-t(C) Cash Flow for Income Taxes	(E)=(A)+(D) ATCF
k	$R_k - E_k$	d_k	$R_k - E_k - d_k$	$-t(R_k - E_k - d_k)$	$(1-t)(R_k - E_k) + td_k$

Example 7.7

- A machine is bought at $180,000
  - Generating revenue $36,000 per year for 10 years
  - Market value is $30,000 at end of the 10th year
- Assume the effective tax rate is 38%
- The machine can be depreciated for 6 years as
  - (36000, 57600, 34560, 20736, 20736, 10368)
- Question
  - (a) Develop the ATCFs and BTCFs.
  - (b) Calculate the before-tax and after-tax IRR.
  - (c) Calculate the after-tax PW when the after-tax MARR = 10% per year.

Example 7.7

Year	(A) BTCF	(B) Depreciation	(C)= (A)-(B) Taxable Income	(D)=-t(C) Cash Flow for Income Taxes	(E)=(A)+(D) ATCF
k	R_k-E_k	d_k	R_k-E_k-d_k	$-t(R_k-E_k-d_k)$	$(1-t)(R_k-E_k) + td_k$
0	-180000				-18000
1	36000	36000	0	0	36000
2	36000	57600	-21600	8208	44208
3	36000	34560	1440	-547	35453
4	36000	20736	15264	-5800	30200
5	36000	20736	15264	-5800	30200
6	36000	10368	25632	-9740	26260
7-10	36000		36000	-13680	22320
10	30000		30000	-11400	18600
Total	210000				Total 130201

Example 7.7

- (b) Calculate the before-tax and after-tax IRR.
  - The before-tax IRR is computed from column A:

  0 =−$180,000 + $36,000(P/A, i%, 10) + $30,000(P/F, i%, 10).
  - By trial and error, we find that i = 16.1%.
  - By trial and error, the after-tax IRR is found to be 12.4%.
- (c) When MARR = 10% per year is inserted into the PW equation, it can be determined that the after-tax PW of this investment is $17,208.

Example 7.7

- IRR
  - Before tax cash flow IRR=16.1%
  - After tax cash flow IRR=12.4%
- If MARR=10%
  - PW before tax cash flow: $52,771
  - PW after tax cash flow: $17,208
- If depreciation for 10 years (details omitted)
  - IRR for after tax cash flow becomes 11.2%
  - PW after tax cash flow: $9,136
- Usually early depreciation is preferred to reduce tax
- Before tax cash flow will not change with depreciation

Example 7.8: compare alternatives

- Option 1: to purchase and use an equipment
  - Capital cost: $20,000
  - Useful life for 7 years with salvage value of $2000
  - Variable operating cost: $40 per day
  - Fixed operating cost: $8,000 per year
  - Depreciation is given for each year
- Option 2: to lease the same equipment
  - Annual cost fixed: $20,000
- Which option is better?
  - To compare Annuity for the cost per year
  - Effective tax rate = 40%
  - MARR=10%

Example 7.8: compare alternatives

- To purchase, suppose the equipment is used for *X* days per year

End of Year, *k*	(*A*) BTCF	(*B*) Depreciation Deduction[a]	(*C*) = (*A*) − (*B*) Taxable Income	(*D*) = −*t*(*C*) Cash Flow for Income Taxes	(*E*) = (*A*) + (*D*) ATCF
0	−$20,000				−$20,000
1	−40X − 8,000	$4,000	−$40X − $12,000	$16X + $4,800	−24X − 3,200
2	−40X − 8,000	6,400	−40X − 14,400	16X + 5,760	−24X − 2,240
3	−40X − 8,000	3,840	−40X − 11,840	16X + 4,736	−24X − 3,264
4	−40X − 8,000	2,304	−40X − 10,304	16X + 4,122	−24X − 3,878
5	−40X − 8,000	2,304	−40X − 10,304	16X + 4,122	−24X − 3,878
6	−40X − 8,000	1,152	−40X − 9,152	16X + 3,661	−24X − 4,339
7	−40X − 8,000	0	−40X − 8,000	16X + 3,200	−24X − 4,800
7	2,000		2,000	−800	1,200

[a] Depreciation deduction$_k$ = $20,000 × (GDS recovery rate).

Example 7.8: compare alternatives

- Calculate the Annuity for the cost with respect to after tax cash flow
  - Note that MARR=10%

$$
\begin{aligned}
AW = &-20000(A/P, 10\%,7) - 24X \\
&-[3200(P/F,10\%,1)+\ldots+4800(P/F,10\%,7)] \\
&\quad (A/P,10\%,7) \\
&+1200(A/F,10\%,7) \\
= &-24X - 7511
\end{aligned}
$$

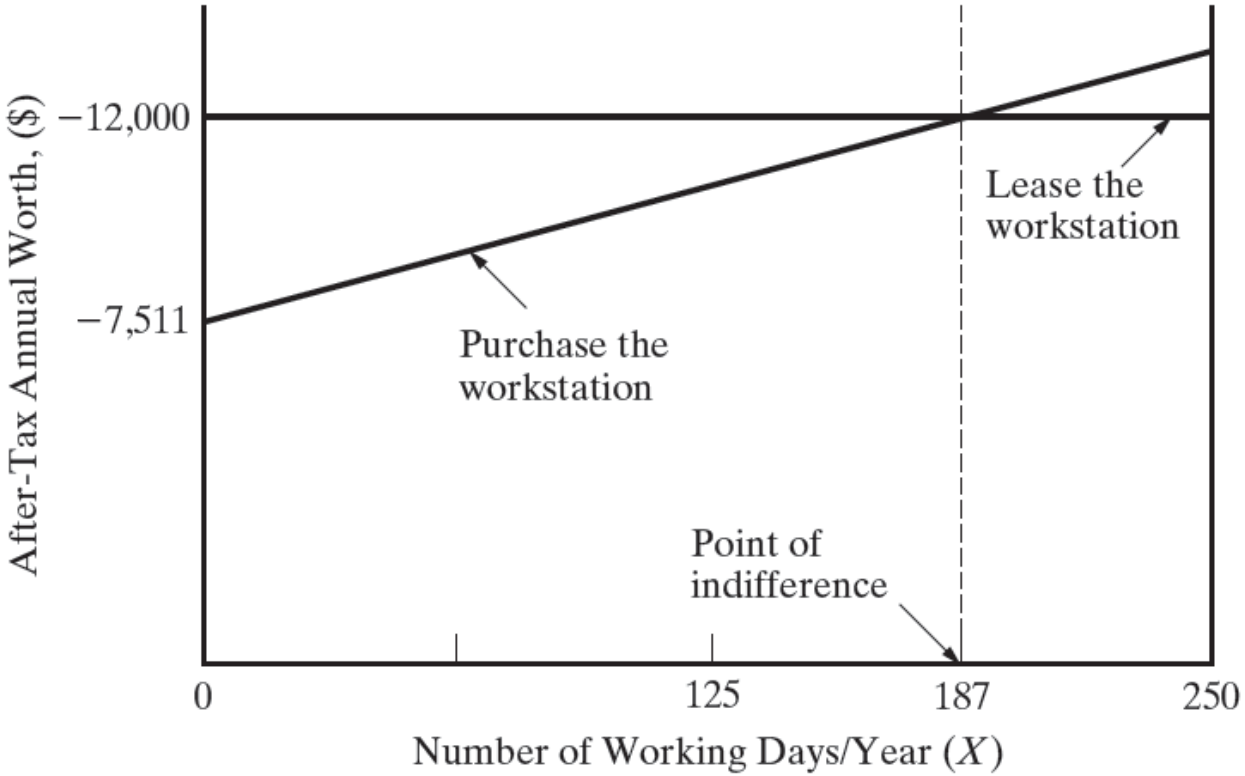
Example 7.8: compare alternatives

- To lease
  - Annual cost is $20,000
  - After 40% tax saving, this is equivalent to

    20000*60%=$12,000 after-tax annual cost
- To purchase
  - Annual cost is 24X+7511
- Breakeven point is 24X+7511=12000

  X = 187 days
- Leasing is better if the equipment is used for more than 187 days

Example 7.8: compare alternatives



Comparison for BTCF

- Without considering tax
  - Annual cost for leasing is 20,000
  - Annual cost for purchasing

    $20000(A/P,10\%,7)+8000-2000(A/F,10\%,7)+40X$

  $=11897+40X$
- Breakeven point

  $11897+40X=20000$

  $X=202.57$

- Chapter 7 of the course textbook.